Distributions and Maximum Likelihood

ANSWER KEY

Section 1: Discrete Random Variables and Bernoulli Likelihood

1 In a binary click-through model, an ad is clicked $k = 40$ times out of $n = 200$ impressions. What is the Maximum Likelihood Estimate $\hat{p}_{MLE}$ of the click probability?

a. 0.20 ✓

b. 0.25

c. 0.16

d. 0.40

For Bernoulli trials, $\hat{p}_{MLE} = k/n = 40/200 = 0.20$.

2 What is the variance of a Bernoulli random variable with success probability $p = 0.3$?

a. 0.21 ✓

b. 0.30

c. 0.70

d. 0.49

$\text{Var}(X) = p(1-p) = 0.3(0.7) = 0.21$.

3 Show how maximizing the Bernoulli log-likelihood relates to the binary cross-entropy loss function used in neural networks.

a. **Maximizing Bernoulli log-likelihood is mathematically identical to minimizing binary cross-entropy** ✓

b. Cross-entropy is the reciprocal of the log-likelihood

c. They only match when $p = 0.5$

d. Cross-entropy ignores negative classes

Log-likelihood is $\sum[y_i \ln p + (1-y_i)\ln(1-p)]$. Binary cross-entropy is the negative log-likelihood $-\text{LL}$.

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Section 2: Continuous Gaussian Normal Distributions

4 A feature vector component follows $X \sim \mathcal{N}(50, 16)$. What are its mean μ and standard deviation σ?

a. $\mu = 50, \quad \sigma = 4$ ✓

b. $\mu = 50, \quad \sigma = 16$

c. $\mu = 16, \quad \sigma = 50$

d. $\mu = 25, \quad \sigma = 4$

Notation $\mathcal{N}(\mu, \sigma^2) \Rightarrow \mu = 50$, $\sigma^2 = 16 \Rightarrow \sigma = 4$.

5 By the empirical $(68 - 95 - 99.7)$ rule, approximately what percentage of samples from $\mathcal{N}(\mu, \sigma^2)$ fall within the interval $(\mu - 2\sigma, \mu + 2\sigma)$?

a. 95.4% ✓

b. 68.2%

c. 99.7%

d. 50.0%

Approximately 95.4% of Gaussian mass lies within ± 2 standard deviations.

6 Standardize the value $x = 62$ from the distribution $\mathcal{N}(50, 16)$ into a standard normal z-score.

a. $z = 3.0$ ✓

b. $z = 0.75$

c. $z = 12$

d. $z = 1.5$

$z = \frac{x - \mu}{\sigma} = \frac{62 - 50}{4} = \frac{12}{4} = 3.0$.

Section 3: The Likelihood Function and Log-Likelihood Formulation

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7 Why is the log-likelihood $\ln L(\theta) = \sum_{i=1}^{n} \ln p(x_i|\theta)$ preferred over raw likelihood in machine learning optimization?

a. **It transforms products of probabilities into sums, preventing numerical underflow and simplifying derivatives ✓**

b. It converts non-convex problems into concave problems unconditionally

c. It eliminates the need for normalization

d. It ensures the sample mean is zero

Multiplying many small probabilities causes float underflow; summing logs is numerically stable and derivatives split cleanly.

8 For n independent samples from $\mathcal{N}(\mu, \sigma^2)$, what is the analytical expression for the log-likelihood?

a. $-\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$ ✓

b. $-\frac{n}{2}\ln(2\pi) - \sum_{i=1}^{n}(x_i-\mu)$

c. $\frac{n}{2\sigma^2}\sum(x_i-\mu)^2$

d. $-\frac{1}{\sigma^2}\sum(x_i-\mu)$

Taking natural log of $\prod \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$ gives this standard form.

9 Given 5 observations $\{2, 4, 4, 6, 9\}$, compute the sample mean $\bar{x}$, which is the Gaussian MLE $\hat{\mu}_{MLE}$.

a. 5.0 ✓

b. 4.0

c. 4.6

d. 5.5

$\bar{x} = \frac{2+4+4+6+9}{5} = \frac{25}{5} = 5.0.$

Section 4: Properties of Estimators: Bias and Consistency

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10 Why is the Maximum Likelihood Estimator for variance $\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(x_i - \bar{x})^2$ biased for finite samples?

a. **Its expected value is $\frac{n-1}{n}\sigma^2 < \sigma^2$, underestimating true variance due to using sample mean $\bar{x}$** ✓

b. It only works for standard normal data

c. Its expected value approaches infinity

d. It ignores outliers

Estimating $\bar{x}$ uses 1 degree of freedom, resulting in $E[\hat{\sigma}^2_{MLE}] = \frac{n-1}{n}\sigma^2$ (Bessel's correction uses $n - 1$).

11 What is Bessel's correction to make the sample variance estimator unbiased?

a. **Multiply by $\frac{n}{n-1}$** ✓

b. Multiply by $\frac{n-1}{n}$

c. Add $\frac{1}{n}$

d. Subtract the mean

$s^2 = \frac{n}{n-1}\hat{\sigma}^2_{MLE} = \frac{1}{n-1}\sum(x_i - \bar{x})^2$.

12 What does it mean for an estimator $\hat{\theta}_n$ to be asymptotically consistent?

a. **As sample size $n \to \infty$, $\hat{\theta}_n$ converges in probability to the true parameter θ** ✓

b. The estimator has zero variance for all sample sizes

c. The estimator is unbiased for $n = 1$

d. The estimator follows a student-t distribution

Consistency is the property that with infinite data, the estimator converges to the ground truth parameter value.

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